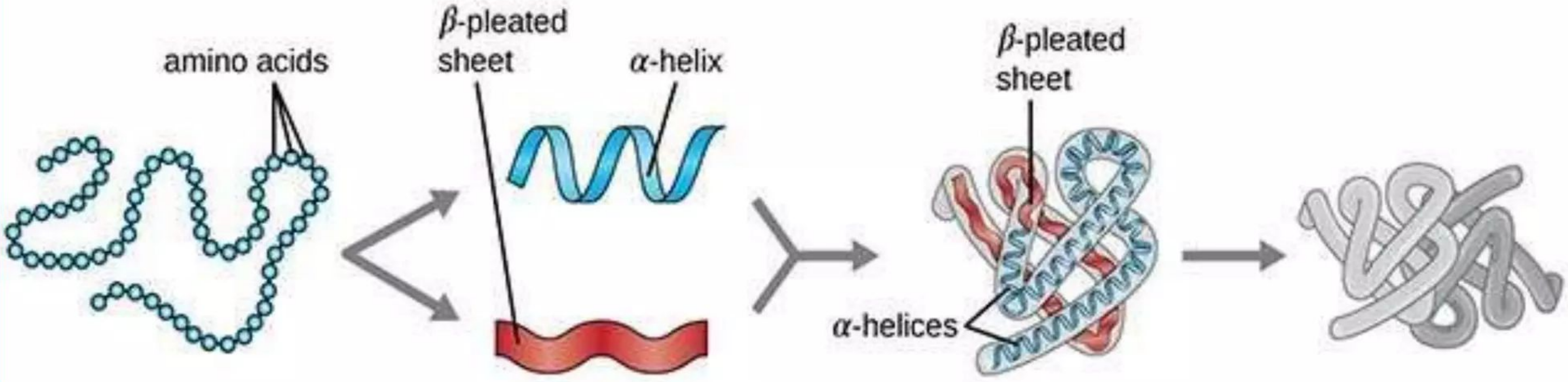
Structure of Proteins



Primary Protein Structure
Sequence of a chain of amino acids

Secondary Protein Structure
Local folding of the polypeptide chain into helices or sheets

Tertiary Protein Structure
three-dimensional folding pattern of a protein due to side chain interactions

Quaternary Protein Structure
protein consisting of more than one amino acid chain

Lecture Objectives:

- By the end of this lecture, students should be able to:
- **Define protein structure** and explain the relationship between amino acid sequence.
- **Describe the four levels of protein structure**:
  - **Primary structure** – amino acid sequence and peptide bonds
  - **Secondary structure** – α-helix and β-pleated sheet formation
  - **Tertiary structure** – overall 3D folding and side-chain interactions
  - **Quaternary structure** – multi-subunit organization
- **Explain the types of bonds and interactions** stabilizing protein structure.
- **Identify common secondary structural elements** such as:
  - α-helices
  - β-sheets
  - Turns and loops

Protein Structure

- Firstly, there is a string of AA's bound with peptide bonds.
- Once the string of AA's interacts with itself and its environment (often aqueous), then we have a functional protein that consists of one or more polypeptides precisely twisted, folded, and coiled into a unique shape.
- The sequence of amino acids determines a protein's three-dimensional structure
- A protein's structure determines its function.

STRUCTURAL ORGANIZATION OF PROTEINS

- Proteins have different level of organization;
- Primary structure: The linear sequence of amino acids forming the backbone of proteins.
- Secondary structure: The spatial arrangement of protein by twisting of the polypeptide chain. Found in most proteins, consists of coils and folds in the polypeptide chain

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- Tertiary structure: The three dimensional structure of a functional protein, determined by interactions among various side chains (R groups).

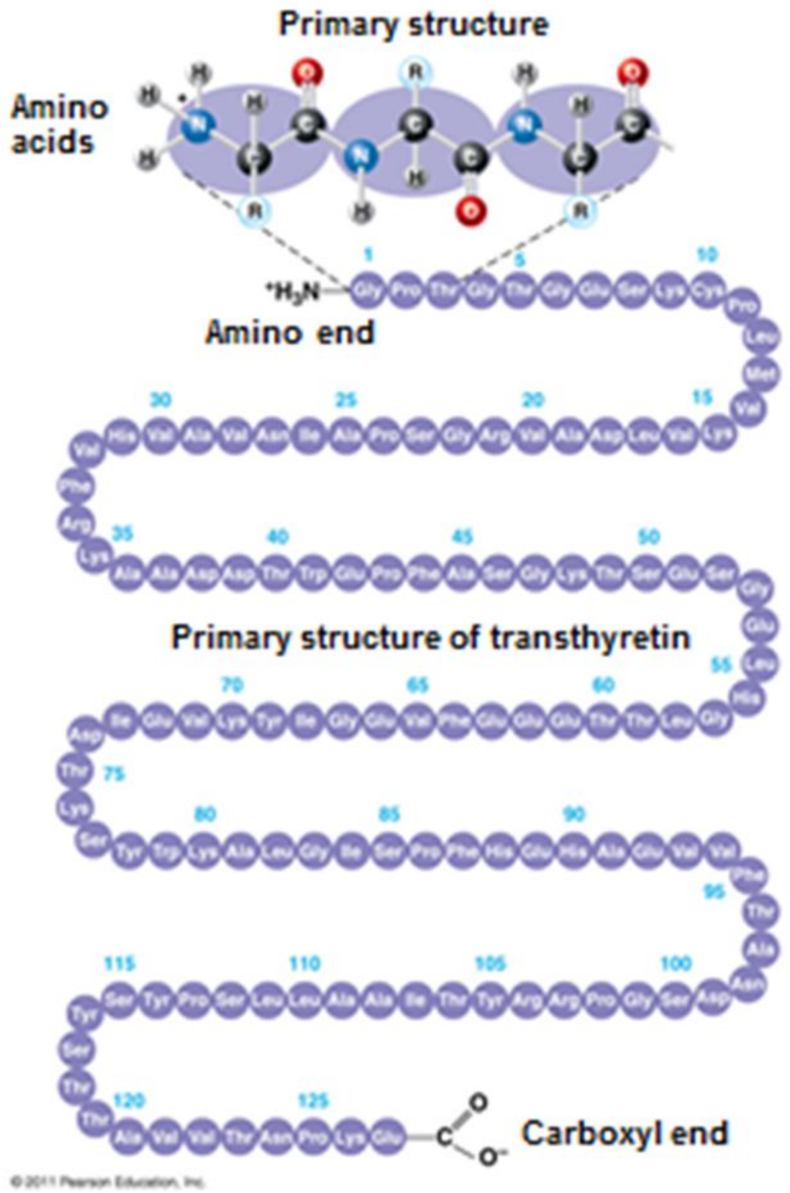
- Quaternary structure: Some of the proteins are composed of two or more polypeptide chains referred to as subunits. The spatial arrangement of these subunits is known as quaternary structure.

Primary structure:

- The linear sequence of amino acids held together by peptide bonds in its peptide chain.

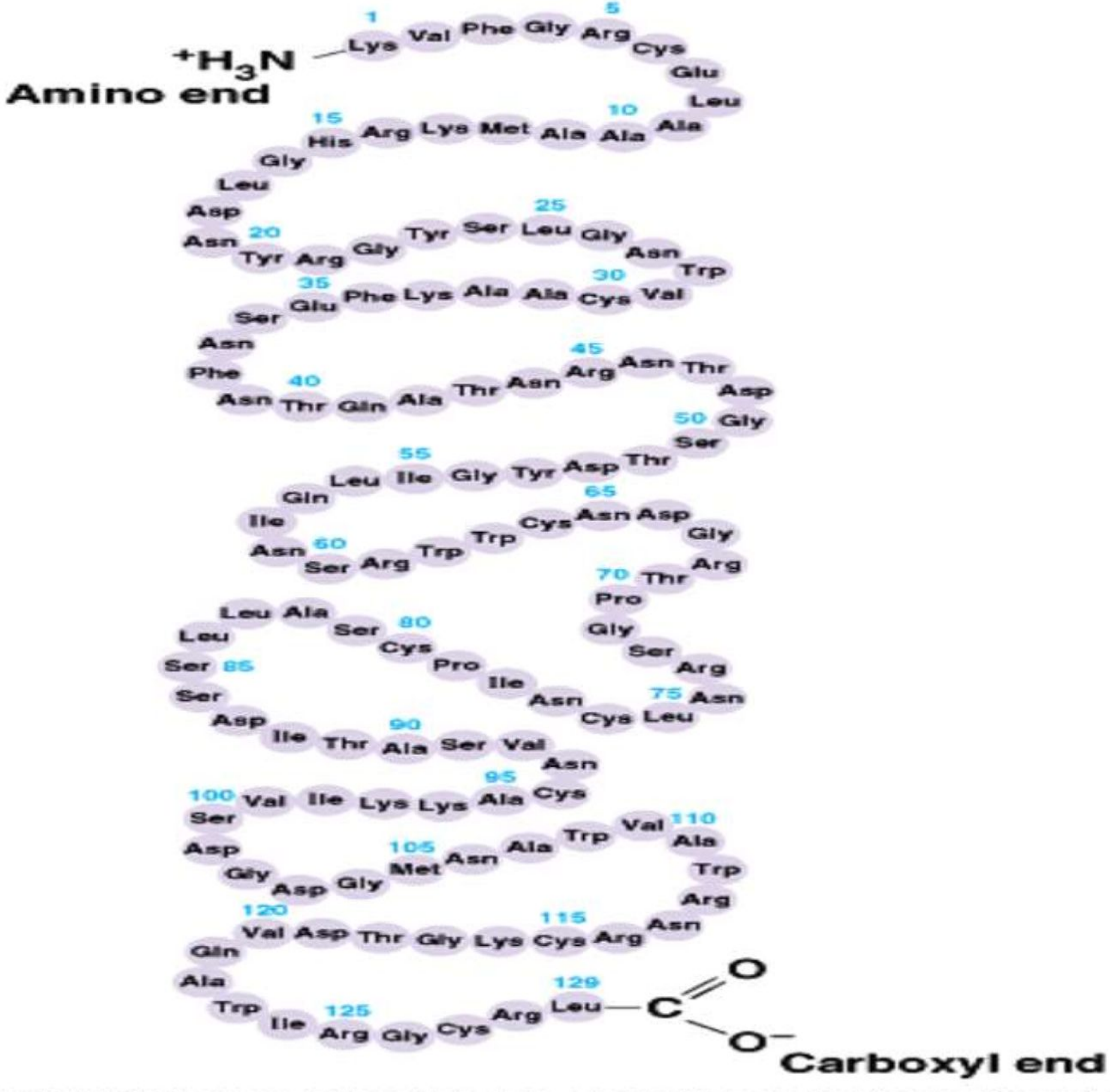
- The peptide bonds form the back bone.

- The free $–NH_2$ group of the terminal amino acid is called as N-terminal end and the free –COOH end is called C-terminal end.



- It is a tradition to number the amino acids from N-terminal end as No.1 towards the C-terminal end.

- Presence of specific amino acids at a specific number is very significant for a particular function of a protein.



Determination of primary structure involves 3 stages:

- Determination of amino acid composition.
- Degradation of protein or polypeptide into smaller fragments.
- Determination of the amino acid sequence

Separation and estimation of amino acids

- The mixture of amino acids liberated by protein hydrolysis can be determined by chromatographic techniques.

Breakdown of polypeptides into fragments

- Polypeptides are degraded into smaller peptides by enzymatic or chemical methods.

- Enzymatic cleavage: The proteolytic enzymes such as *trypsin, chymotrypsin, pepsin and elastage* exhibit specificity in cleaving the peptide bonds.

- Among these enzymes, trypsin is commonly used.

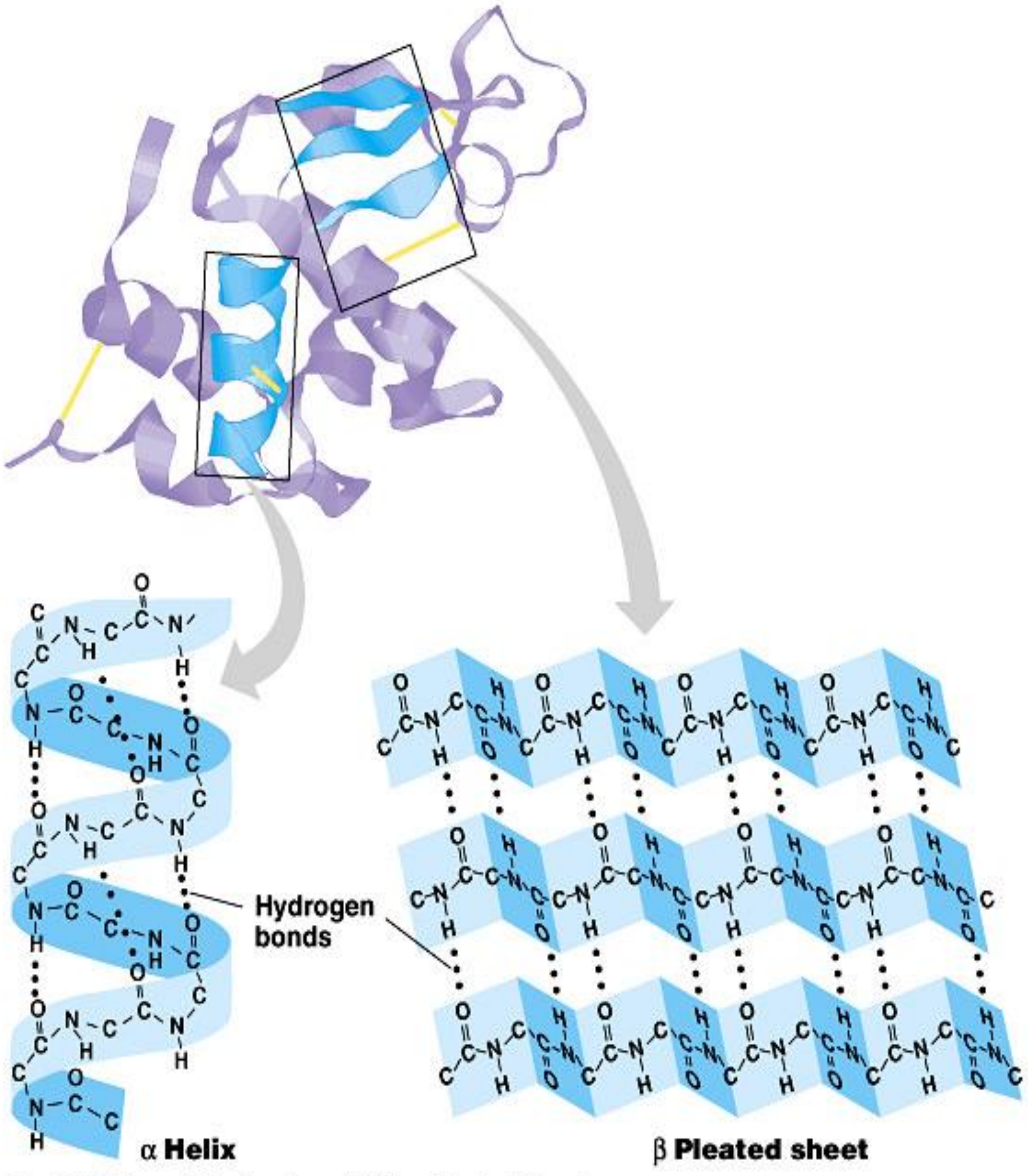
Reverse sequencing technique

- It is the genetic material (DNA) which ultimately determines the sequence of amino acids in a polypeptide chain.
- By analyzing the nucleotide sequence of DNA that codes for protein, it is possible to translate the nucleotide sequence into amino acid sequence.
- This technique, however, fails to identify the disulfide bonds and changes that occur in the amino acids after the protein is synthesized.

Secondary Structure

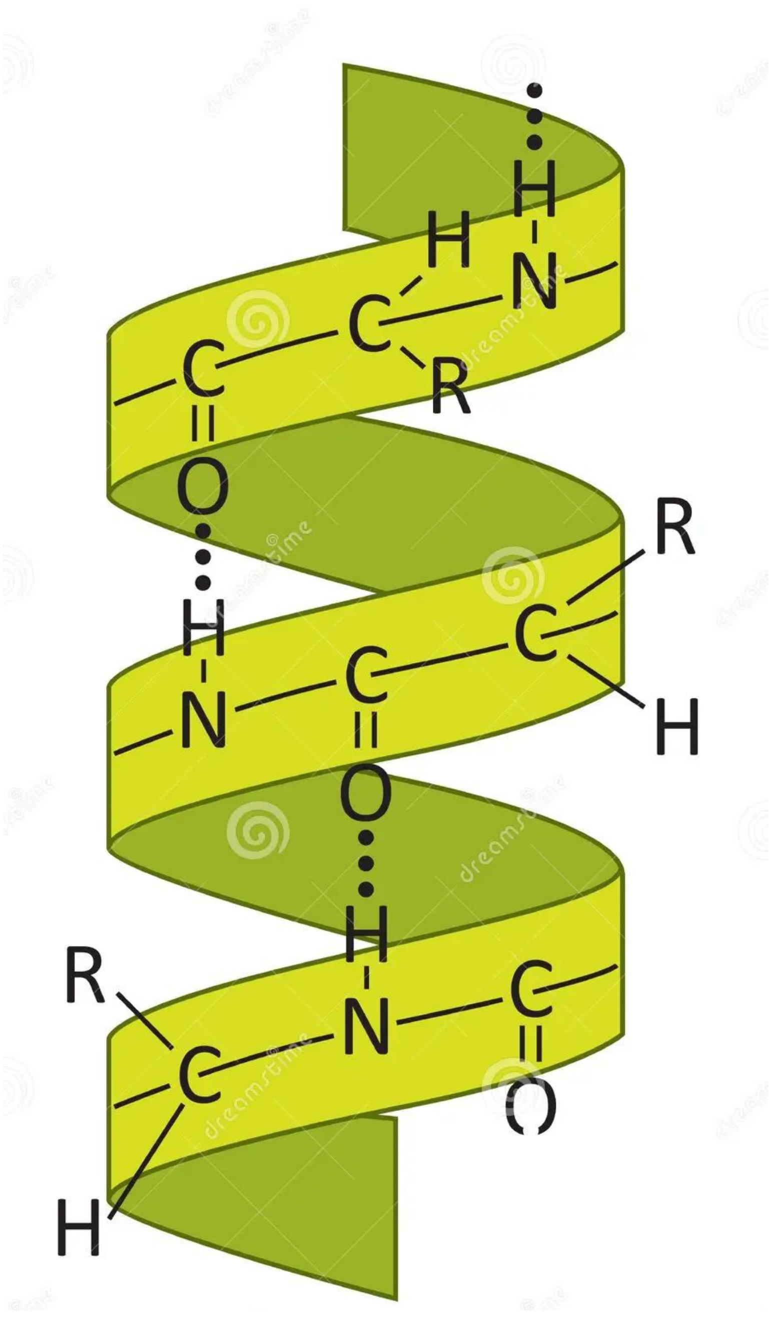
- The coils and folds of **secondary structure** result from **hydrogen bonds** between repeating constituents of the polypeptide backbone.

- Typical secondary structures are a coil called an **α helix** and a folded structure called a **β pleated sheet**



α-Helix

- α-Helix is the most common spiral structure of protein.
- The α-Helix is a tightly packed coiled structure with amino acid side chains extending outward from the central axis.
- The α-Helix is stabilized by extensive hydrogen bonding
- Formed between H atom attached to N, and O atom attached to C.



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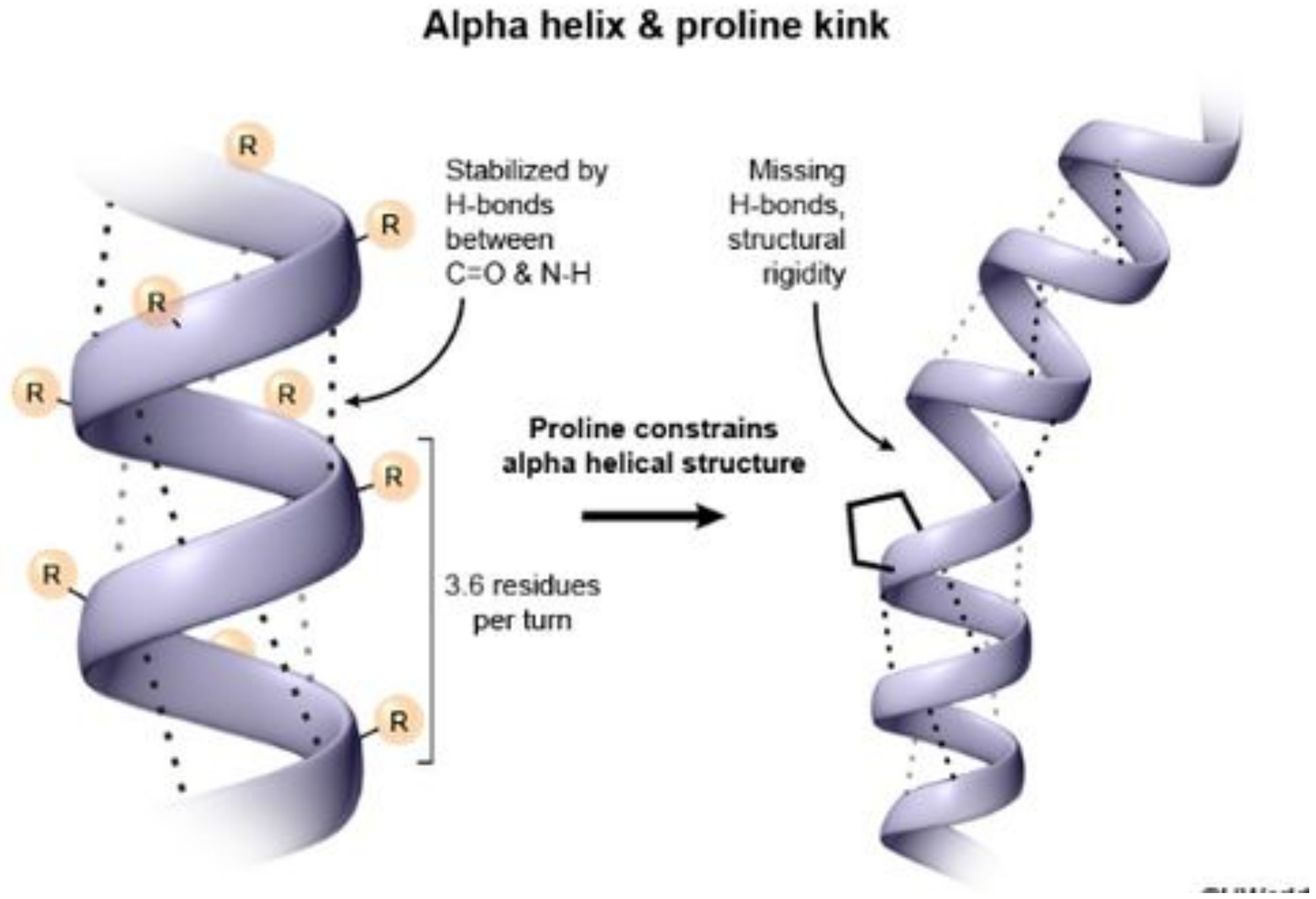
- The H-bonds are individually weak but collectively, strong enough to stabilize the helix.
- All the peptide bonds, except the first and last in polypeptide chain, participate in hydrogen bonding.
- Each turn of helix contains 3.6 amino acids and travels a distance of 0.54 nm.
- The right handed α-Helix is more stable than the left handed helix

α-helix

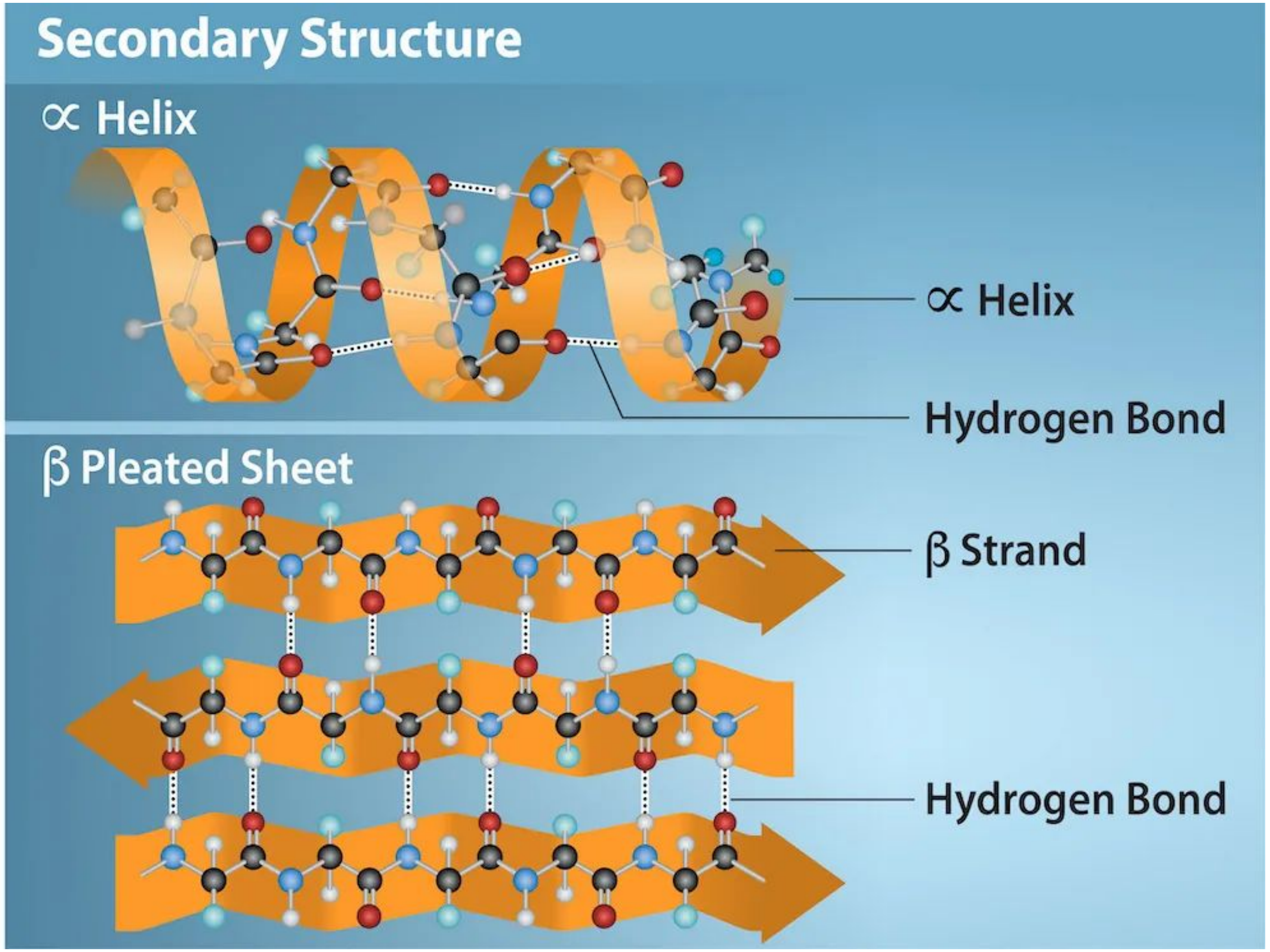
- Certain amino acids (particularly proline) disrupt the α-Helix.

- Large number of acidic and basic amino acids are also interfere with α-Helix structure.



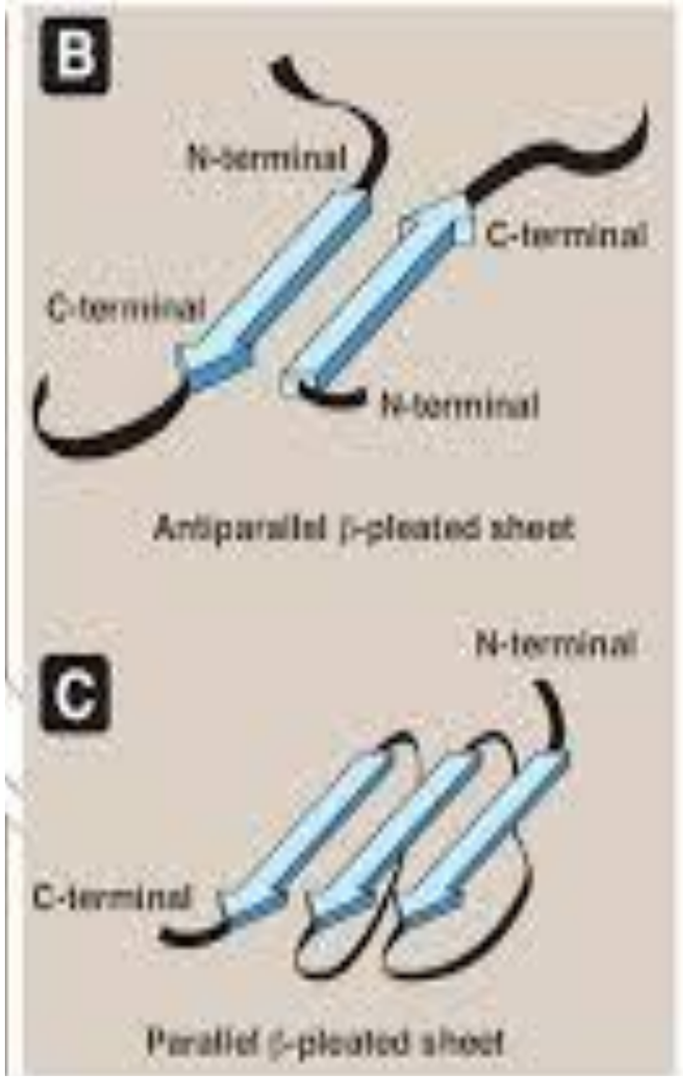
β-pleated sheet

- An another form of secondary structure in which two or more polypeptides (or segments of the same peptide chain) are linked together by hydrogen bond between H- of NH- of one chain and carbonyl oxygen of adjacent chain (or segment).



- The polypeptide chains in the β-sheets may be arranged either in parallel or anti-parallel.

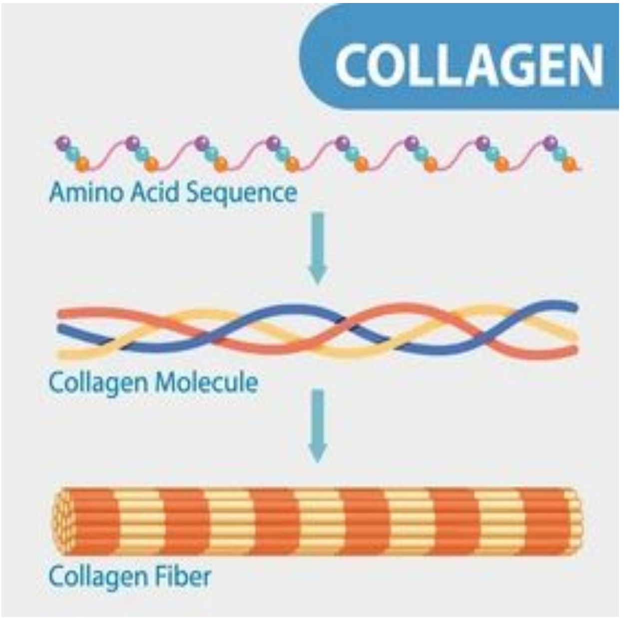
- β-Pleated sheet may be formed either by separate polypeptide chains (H-bonds are inter chain) or a single polypeptide chain folding back on to itself (H-bonds are intrachain)



Triple helix

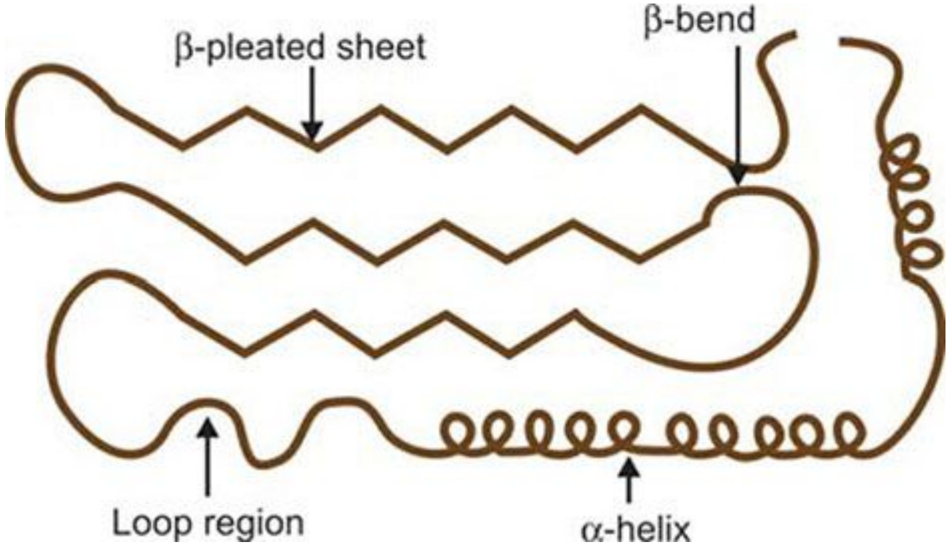
- Collagen is rich in proline and hydroxy proline and cannot form a α-helix or β-Pleated sheet.

- Collagen forms a triple helix.
- The triple helix is stabilized by both non covalent as well as covalent bonds.



Reverse Turns or β-bends

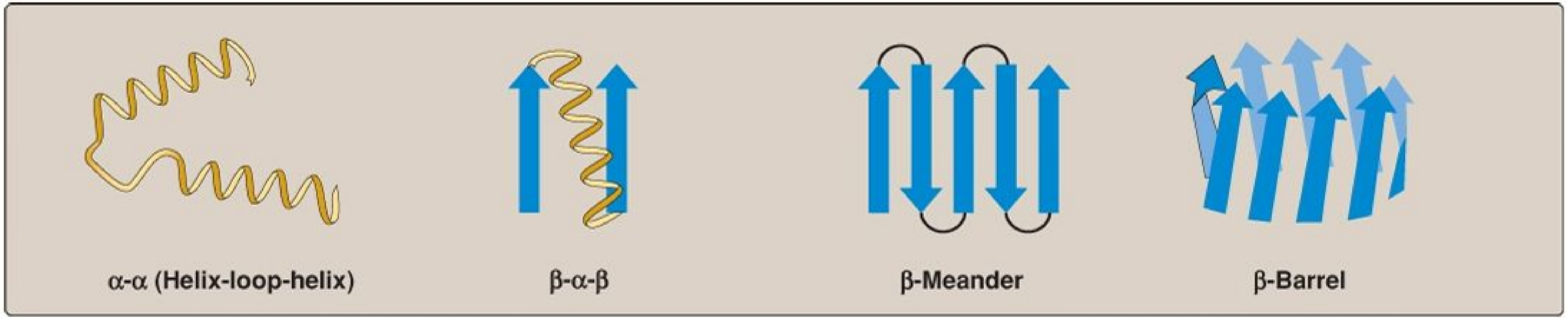
- Since the polypeptide chain of a globular protein changes direction two or more times when it folds, the conformation known as Reverse turns or β-bends.
- Reverse turns usually occur on the surfaces of globular proteins.



- Super secondary structures
- Varies combinations of secondary structure, called super secondary structure, are commonly found in globular proteins.

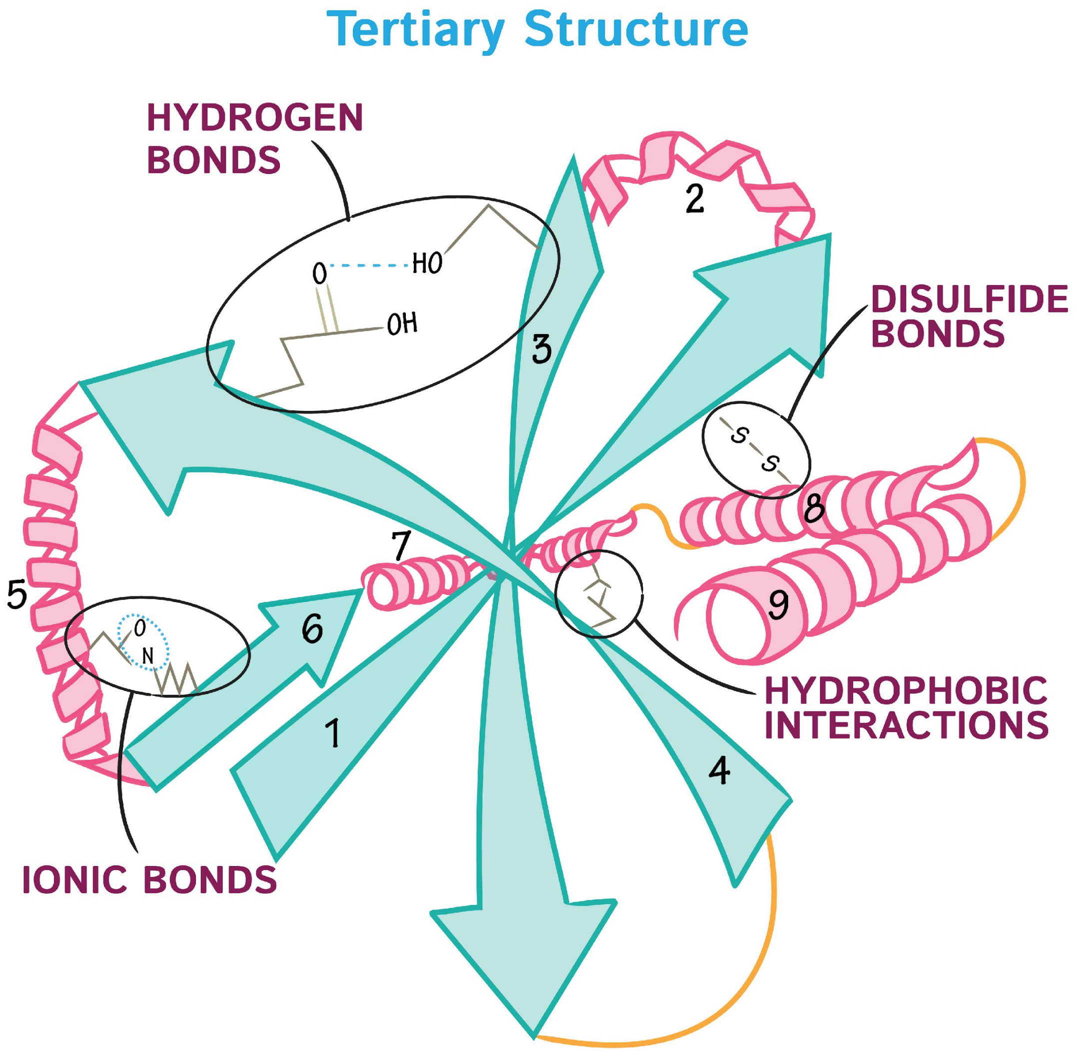
- β- α- β unit : The β- α- β unit consists of two parallel β- pleated sheets connected by an intervening strand of α –helix.

- β-meander :The β-meander consists of five β-Pleated sheets connected by reverse turns.



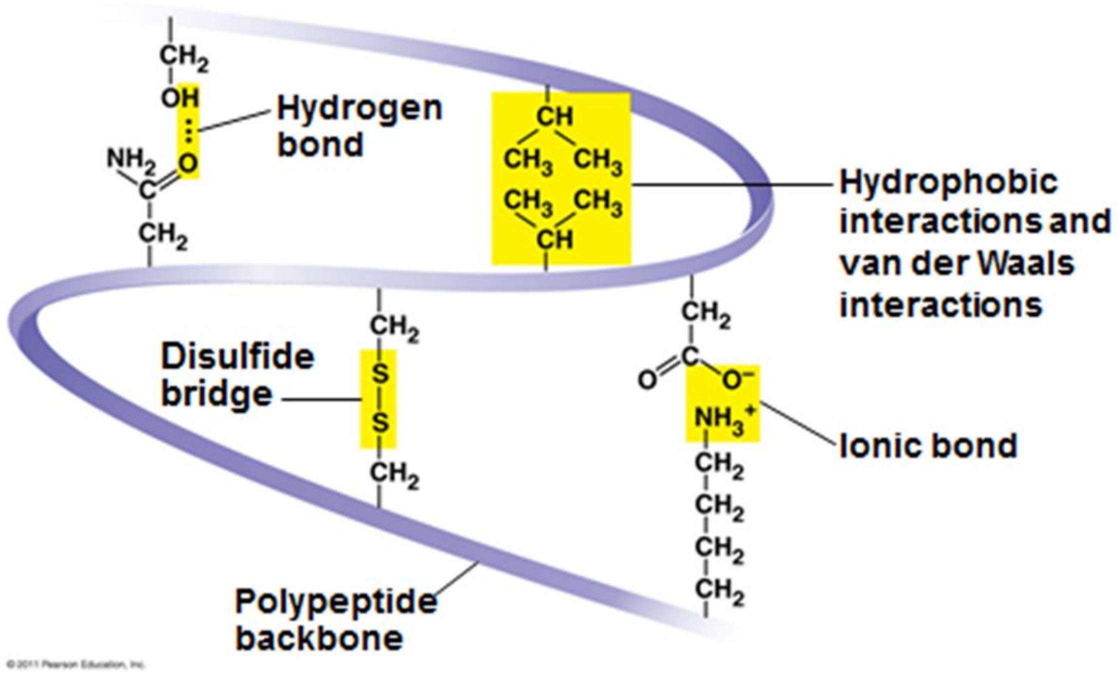
Tertiary structure

- The three dimensional structure of a protein.
- It is a compact structure with hydrophobic side chains held interior while the hydrophilic groups are on the surface of the protein.
- This type of arrangement ensures stability of the molecule.



Bonds of Tertiary structure

- The hydrogen bonds,
- Disulfide bonds(-S-S)
- Ionic interactions (electrostatic bonds) and
- Hydrophobic interactions also contribute to the tertiary structure of proteins.



- Domains: Used to represent the basic unit of protein structure (tertiary) and function.

- A polypeptide with 200 amino acids normally consists of two or more domains.

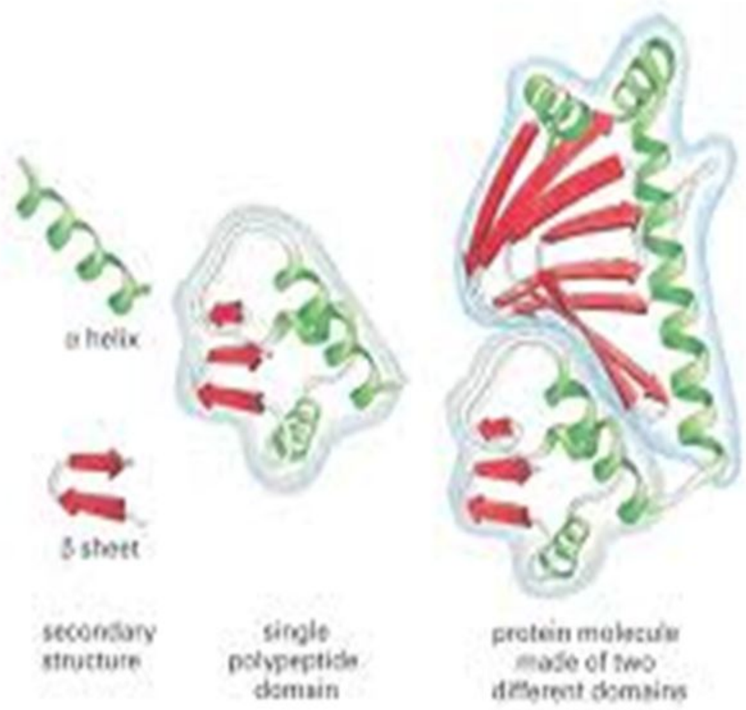
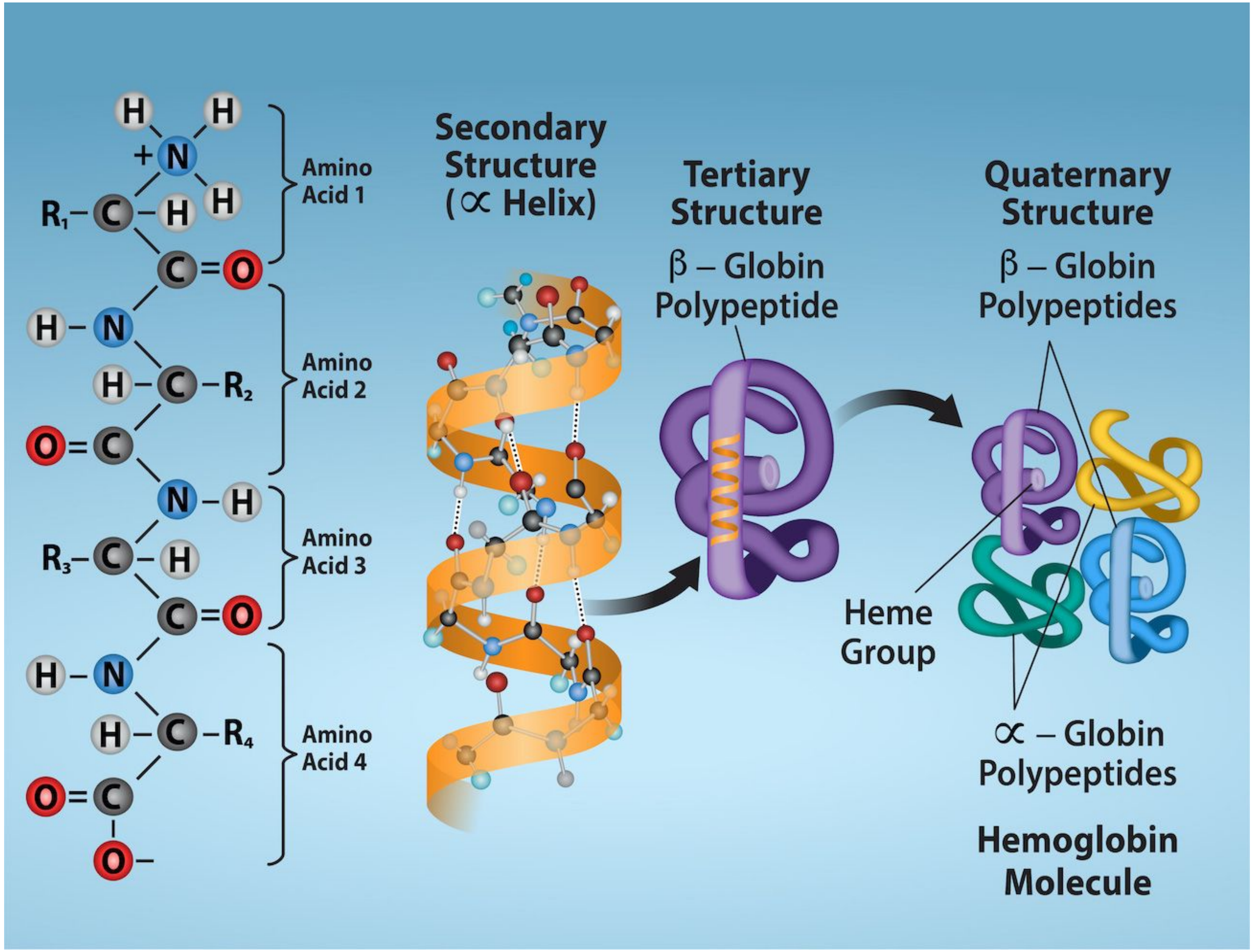


Figure 4-19 Essential Cell Biology, 2/e. (© 2004 Garland Science)

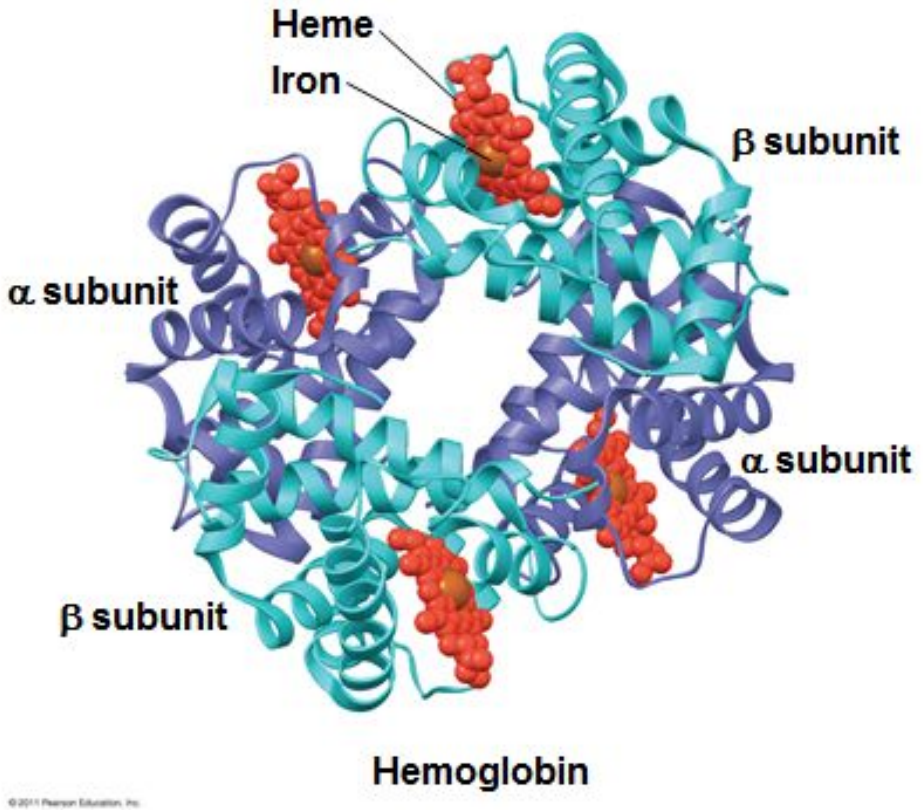
Quaternary structure

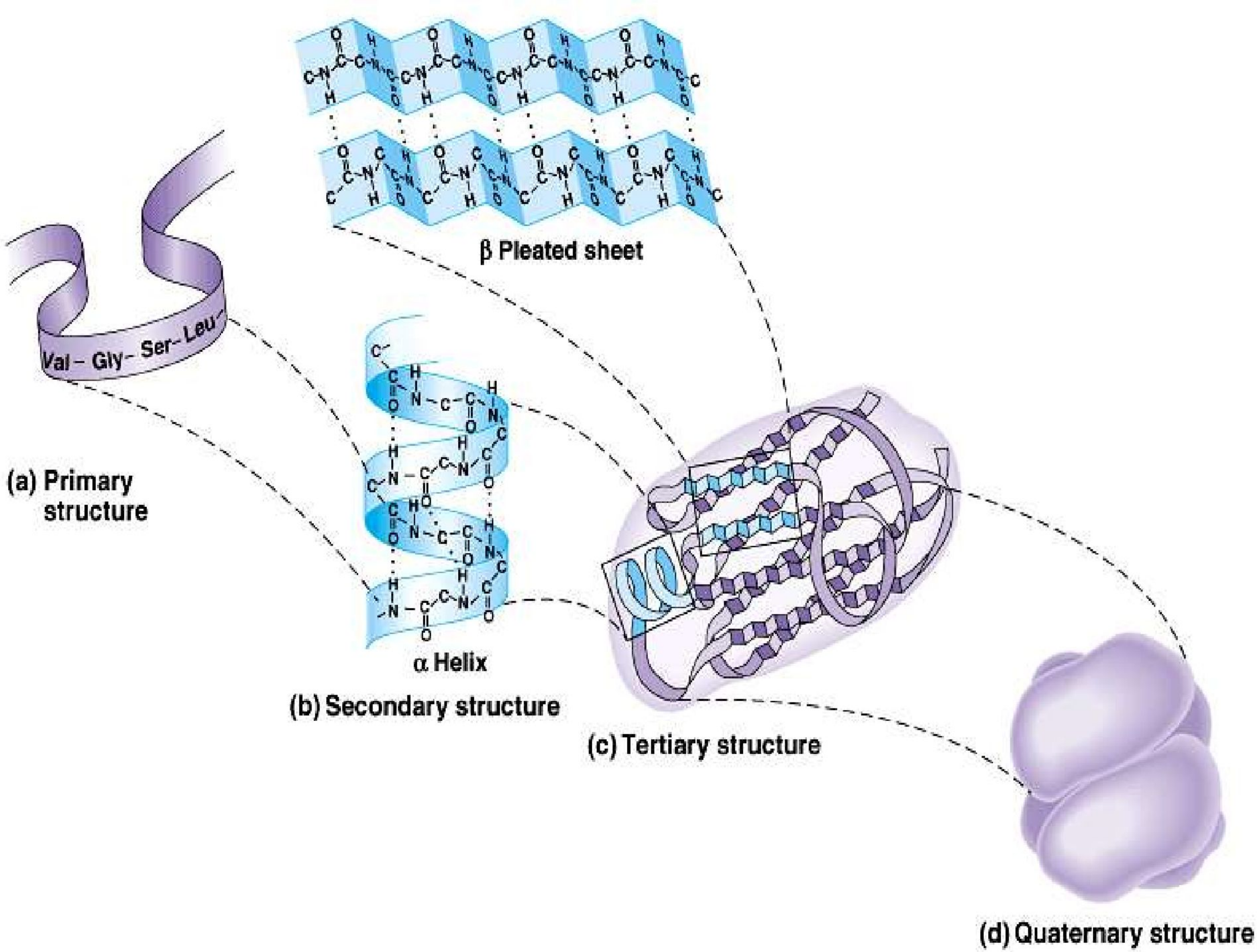
- Majority of the proteins are composed of single polypeptide chains.
- Some of the proteins, consists of two or more polypeptides which may be identical or unrelated.
- Such proteins possess quaternary structure



Quaternary Structure

- Hemoglobin is a globular protein consisting of four polypeptides: two alpha and two beta chains




β Pleated sheet
Val – Gly– Ser– Leu
(a) Primary structure
α Helix
(b) Secondary structure
(c) Tertiary structure
(d) Quaternary structure

Bonds responsible for protein structure

- Two types of bonds

  Covalent and non-covalent bonds

<u>Covalent bonds:</u>

- The peptide and disulfide bonds are the strong bonds in protein structure.

Disulfide bond:

- A disulfide bond (-S-S-) is formed by the sulfhydryl groups groups (-SH) of two cysteine residues to produce cystine.

Non-covalent bonds

Hydrogen bonds (H-bonds):

- The H-bonds are formed by sharing of H-atoms between the Nitrogen and carbonyl oxygen of different peptide bonds.
- Each H-bond is weak but collectively they are strong.
- A large number of H-bonds significantly contribute to the protein structure.

Hydrophobic bonds

- The non-polar side chains of neutral amino acids tend to be closely associated with each other in proteins.

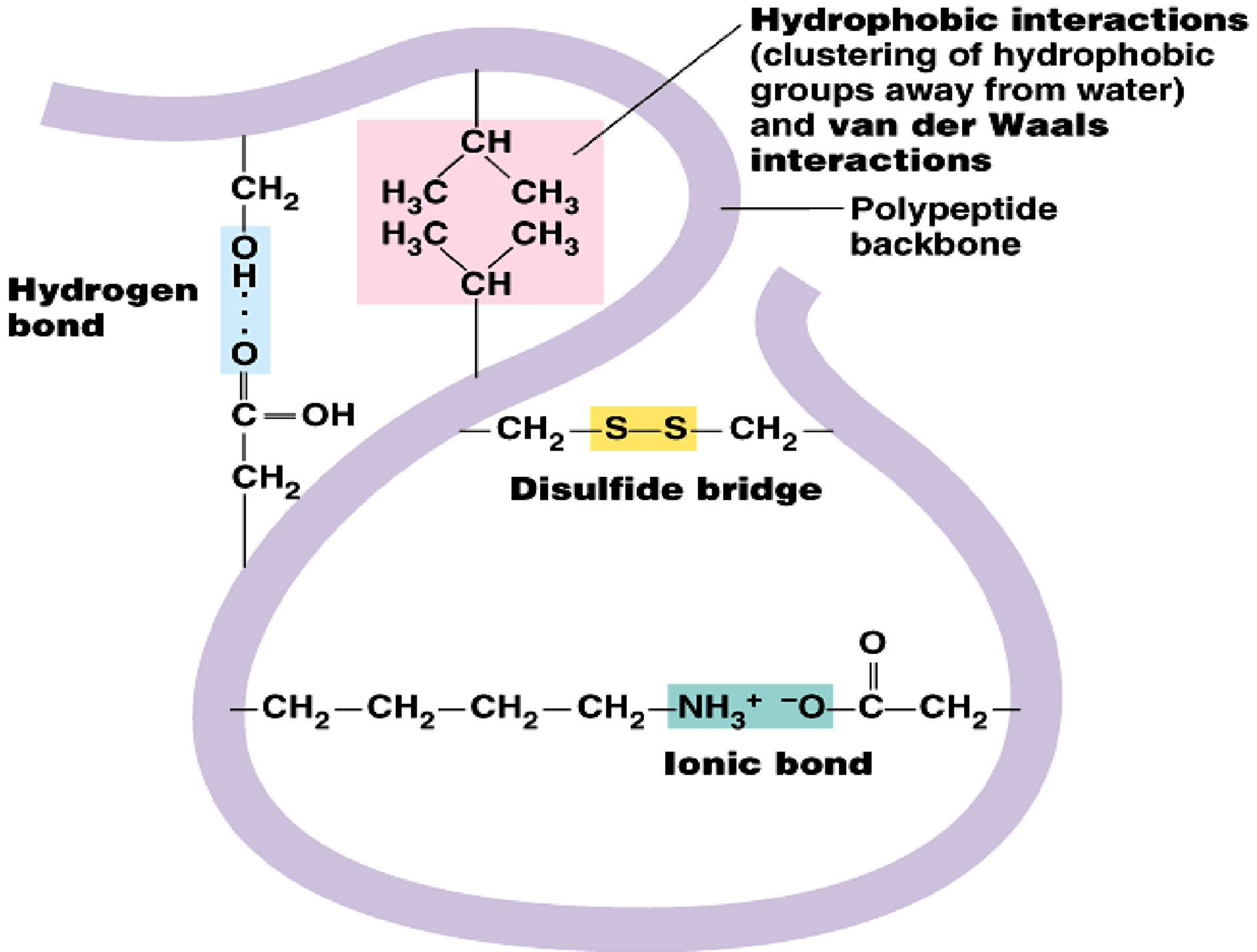
- These not true bonds.
- The occurrence of hydrophobic forces is observed in aqueous environment wherein the molecules are forced to stay together.

Electrostatic bonds:

- These bonds are formed by interactions between negatively charged groups of acidic amino acids with positively charged groups of basic amino acids.

Van der Waals forces:

- These are the non-covalent associations between electrically neutral molecules.



Hydrophobic interactions (clustering of hydrophobic groups away from water) and van der Waals interactions
Polypeptide backbone
Hydrogen bond
Disulfide bridge
Ionic bond
CH
H3C
CH3
CH2
O
H
C
OH
S
NH3+
-O

Thank you